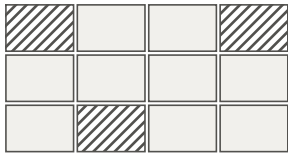


a

tiles of one section



 training tiles: fit the baseline and the probe

 held-out tiles: score both

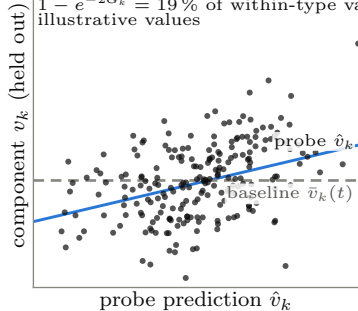
baseline: $\bar{v}_k(t)$, the mean over cells of type t

probe: $\hat{v}_k(\boldsymbol{\mu}_z, t)$, ridge or MLP

b

$$G_k = \frac{1}{2} \log \frac{\text{MSE}_{\text{type}}}{\text{MSE}_{\text{probe}}} = 0.10 \text{ nats}$$

$1 - e^{-2G_k} = 19\%$ of within-type variance
illustrative values

**c**

 floor  this fit  uncontrolled

composition



fraction of uncontrolled = 0.31

image



fraction of uncontrolled = 0.16

G_b (nats per component), illustrative